C10 — A year forward on Bybit, the crash, and data from outside the book

Not a ladder phase. It answers the first two of the "next things to try" after C9: a forward test of the frozen C8 choices on data they have never seen, and inputs from outside the traded book. Lab notebook: `experiments/c10_forward/README.md`.

> **At a glance**
>
> - **The C8 edge does not survive a year on Bybit.** Six frozen OBI configurations, run unchanged on one Bybit session a week from 2025-09-26 to 2026-10-02 (54 sessions): without the 2025-10-10 crash only XRPUSDC → XRPPERP makes money (+3.96%, 30 of 53 sessions up), and it falls to +0.21% at Bybit's standard +2 bp maker fee. At the assumed rebate the other five lose 0.1–5.8%.
> - **The venue alone already costs most of it.** On the study month's own held-out week, replayed on Bybit, the six make +1.41 to −1.37% against +2.5 to +3.5% on Binance.
> - **One session decides the year's headline.** On 2025-10-10 Bybit's SUI fell 82%, AVAX 67% and XRP 54% within minutes. As frozen the six make anything from +34.7% to −143.2% that day, by luck of the inventory held when it began: the position cap was valued at the falling price and ignored resting orders, so the quoter kept buying.
> - **None of C8's risk controls bounds that day; a stop does.** Circuit breakers, daily loss limits and a units cap each leave some configuration with a crash-day drawdown of 26–211%. A loss limit that *exits* the position at the touch (`loss_flatten`) holds every configuration's drawdown to 0.8–6.4% at 50 bp, and is also the cheapest control on the rest of the year (summed −5.6% against −8.4% as frozen). It keeps the crash bounded under every stress (worst drawdown 3.5–6.4% with fees up to +2 bp, latency ×4 and a pessimistic queue); it does not make a losing configuration profitable.
> - **Outside data improves the forecasts, if it arrives at once.** Adding the same asset's Binance perpetual and spot trades and Bybit's perpetual roughly doubles the held-out R^2 of the 1 s move (XRP 6.0 → 8.9%, SUI 2.5 → 6.3%, ETH 3.8 → 8.4%); most of the gain is gone 100 ms later. The large-move classifier gains 2–3 points of precision at any delay. Whether either pays in the backtest: §8.

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1. The task

From the list of next steps after C9, the request was "begin 1 and 2":

1. **A forward test.** Every Part C conclusion rests on seven held-out Binance sessions from September 2025. Run the configurations that made money there, with every parameter frozen, on a year of sessions they have never seen.
2. **Lead-lag inputs.** C9 found that knowing the mid 1–3 s ahead is worth a lot and that the traded book alone forecasts 2–10% of it. Large moves usually start elsewhere: does the same asset's other markets' data forecast them better, and does it still when it arrives late?

The third item, a host to record Binance's own books going forward, is the user's (free-tier options are in the Part C overview).

2. The data: Bybit's free books

Binance publishes trades but not order books; Bybit publishes both, free and without an account (`bybit.py`):

- **books** — `quote-saver.bycsi.com`, 200 levels, a snapshot then deltas pushed every 10–100 ms, one JSON message per line with Bybit's publish time (`ts`) and the matching engine's (`cts`);
- **trades** — `public.bybit.com/trading`, every print with the taker's side and an RPI flag.

They are converted to the same `data/npz/<symbol>_<date>.npz` the Binance days use, so every C6–C9 script runs on them unchanged. Three choices:

- a book event's exchange time is its matching-engine time, its local time the publish time plus 2 ms (the median publish delay was 4–27 ms a day);
- order latency is not taken from that feed delay: batching delays the data, not our orders (a constant 3 ms feed term for Bybit symbols);
- **RPI trades are dropped**: Retail Price Improvement orders trade only with non-API takers and are not in the public book, so a print against one could not have filled our quote (a median 3.9% of prints a day, at most 16.6%).

Each Binance configuration runs on the Bybit contract that matches it:

Binance (C8 choice)	Bybit	held out on Binance, %
XRPUSDC OBI (floor 12 bp, κ 2, skew ×2)	XRPPERP (USDC-settled)	+2.84
XRPUSDT OBI (floor 8 bp, κ 2)	XRPUSDT	+3.53
SUIUSDC OBI (floor 8 bp, κ 4)	SUIPERP (USDC-settled), and SUIUSDT	+2.52
SUIUSDT OBI (floor 16 bp, κ 4)	SUIUSDT	−0.53
AVAXUSDC OBI (floor 16 bp, κ 0)	AVAXUSDT (no USDC AVAX perpetual)	+2.68

Sessions, fixed before any was run: the **venue control** — the study month's held-out week (2025-09-19 to 09-25) replayed on Bybit — and the **forward set**, every 7th day from 2025-09-26 (the day after the study month) to 2026-10-02, 54 sessions.

3. The forward test

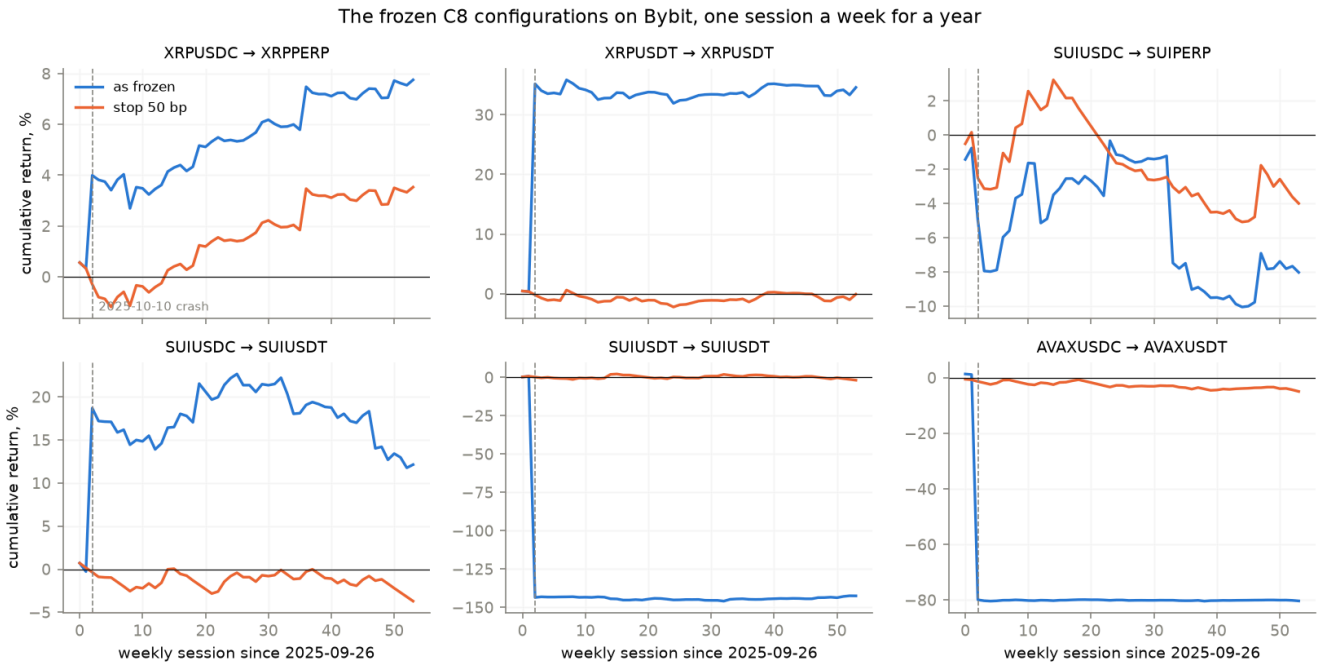


Figure — cumulative return of each frozen configuration over the 54 weekly sessions, as frozen (blue) and with the stop picked in §5 (orange). The dashed line is the 2025-10-10 crash.

configuration	as frozen (−0.5 bp)	fee 0	fee +1 bp	fee +2 bp (Bybit VIP 0)	latency ×4	queue risk-averse	fills/day	sessions up
XRPUSDC → XRPPERP	**+3.96**	+3.20	+1.70	+0.21	+0.52	+3.68	42	30/53
XRPUSDT → XRPUSDT	−0.09	−4.98	−14.09	−22.36	+0.66	+7.01	206	25/53

configuration	as frozen (−0.5 bp)	fee 0	fee +1 bp	fee +2 bp (Bybit VIP 0)	latency ×4	queue risk-averse	fills/day	sessions up
SUIUSDC → SUIPERP	−3.94	−4.90	−6.80	−8.66	−6.18	−3.21	28	28/53
SUIUSDC → SUIUSDT	−5.79	−7.56	−11.01	−14.34	−10.18	−4.58	88	26/53
SUIUSDT → SUIUSDT	−1.65	−2.15	−3.15	−4.15	+2.77	+0.50	32	26/53
AVAXUSDC → AVAXUSDT	−0.92	−2.02	−4.19	−6.32	−0.41	−1.30	46	30/53

Chained return, %, over the 53 forward sessions without 2025-10-10. Fills: median per session, as frozen.

configuration	as frozen (−0.5 bp)	fee 0	fee +1 bp	fee +2 bp	latency ×4	queue risk-averse	fills/day	sessions up
XRPUSDC → XRPPERP	+1.41	+1.30	+1.07	+0.85	+0.69	+1.41	67	4/7
XRPUSDT → XRPUSDT	−1.37	−1.98	−3.19	−4.38	−1.02	−1.82	324	4/7
SUIUSDC → SUIPERP	+0.72	+0.22	−0.77	−1.75	+1.41	+0.46	276	5/7
SUIUSDC → SUIUSDT	−0.52	−0.96	−1.83	−2.70	−0.09	−0.76	327	5/7
SUIUSDT → SUIUSDT	−1.05	−1.10	−1.19	−1.29	−0.95	−0.95	26	4/7
AVAXUSDC → AVAXUSDT	−0.78	−1.22	−2.10	−2.98	+0.39	−0.27	196	3/7

The venue control: the same seven held-out sessions as C8, on Bybit.

Read together:

- **The edge does not survive the venue.** On the very sessions where C8 measured +2.5 to +3.5% on Binance, the same configurations make at most +1.41% on Bybit, and three of six lose. The floors and κ were tuned on Binance's books; why Bybit's differ for these quoters (depth, queue competition, the RPI flow that never reaches a public quote) was not separated here.
- **The year confirms it, with one exception.** XRPUSDC → XRPPERP is the only configuration that makes money on ordinary sessions (+3.96%, 30 of 53 up, positive under every stress but latency ×4 at +0.52%). It is also the one C8 found robust to costs. At Bybit's standard maker fee it is +0.21%: it needs a rebate or a market-maker programme to be worth running.

- **With the crash the picture is upside down.** All 54 sessions: XRPUSDT +34.6%, SUIUSDC → SUIUSDT +12.1% — and SUIUSDT −142.5%, AVAXUSDT −80.3%. Those numbers are one session's luck (§4), not an edge.

4. The crash session

At 21:21 UTC on 2025-10-10 the Bybit books of SUI, AVAX and XRP fell 82%, 67% and 54% within minutes and recovered most of it within the hour (the replay was checked against the raw messages: these prints are real).

configuration	crash-day return, %	max drawdown, %	largest position, $k
XRPUSDC → XRPPERP	+3.6	46.4	164
XRPUSDT → XRPUSDT	+34.7	7.7	84
SUIUSDC → SUIPERP	−4.3	8.9	60
SUIUSDC → SUIUSDT	+19.0	35.1	69
SUIUSDT → SUIUSDT	−143.2	149.1	136
AVAXUSDC → AVAXUSDT	−80.1	124.0	141

As frozen. Capital and the position cap are both $100k.

Losses beyond 100% of capital should be impossible with a $100k cap. They came from how the cap was checked (`zoo.quote_once`): as notional at the *current* mid, so a falling price made room to buy more units; and against the position only, so a whole 20-level resting grid could fill past it at once. The quoter bought all the way down. Two fixes were added to the kernel, off by default (the cached results are unchanged): `cap_ref` values the cap at the session's first mid (a cap in units), and `cap_resting` counts resting orders against it.

5. Risk controls and stops

C8's controls, and the two cap fixes, each ran on every forward session, so their cost on ordinary days is measured with their effect on the crash:

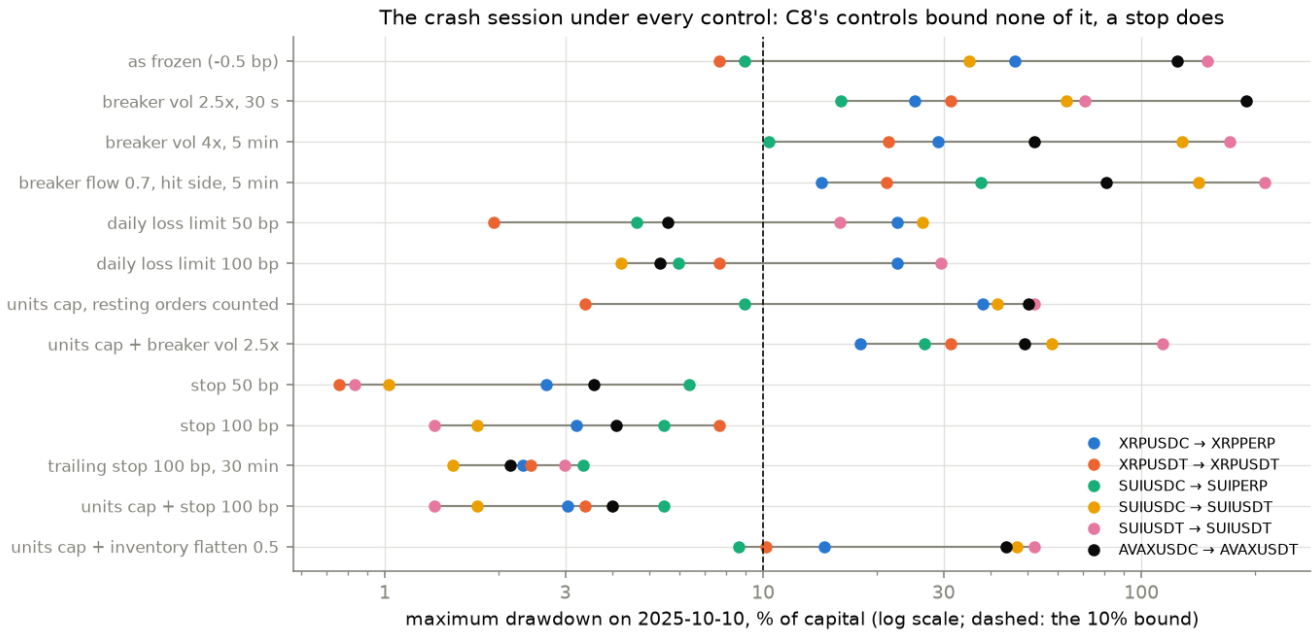


Figure — maximum drawdown on 2025-10-10 under each control, per configuration (log scale). Dashed: the 10% bound of the selection rule.

configuration	as frozen	breaker vol 2.5×, 30 s	breaker vol 4×, 5 min	breaker flow 0.7, hit side, 5 min	daily loss limit 50 bp	daily loss limit 100 bp	units cap, resting counted	units cap + breaker vol 2.5×
XRPUSDC → XRPPERP	+3.6 / 46.4 ; +3.96	+18.3 / 25.1 ; −0.02	+9.8 / 29.1 ; +4.76	+34.1 / 14.3 ; −4.27	−1.1 / 22.6 ; +3.81	−1.1 / 22.6 ; +3.96	−2.6 / 38.0 ; +3.96	+1.3 / 18.0 ; −0.02
XRPUSDT → XRPUSDT	+34.7 / 7.7 ; −0.09	−18.3 / 31.3 ; +3.18	+2.2 / 21.5 ; +0.59	+82.6 / 21.2 ; −1.93	+2.5 / 1.9 ; +1.27	+34.7 / 7.7 ; −0.92	+15.6 / 3.4 ; −0.07	−1.3 / 31.3 ; +3.59
SUIUSDC → SUIPERP	−4.3 / 8.9 ; −3.94	−12.6 / 16.0 ; +6.17	+2.2 / 10.4 ; +3.61	+20.7 / 37.7 ; +1.28	+0.0 / 4.6 ; −3.01	−1.9 / 6.0 ; −4.68	−4.3 / 8.9 ; −3.24	−21.0 / 26.6 ; +5.89
SUIUSDC → SUIUSDT	+19.0 / 35.1 ; −5.79	+241.5 / 63.3 ; −10.52	−121.7 / 128.2 ; −4.20	−95.5 / 141.8 ; −6.70	−24.7 / 26.4 ; −5.85	−3.3 / 4.2 ; −6.56	−40.0 / 41.5 ; −5.12	−55.0 / 58.0 ; −10.38
SUIUSDT → SUIUSDT	−143.2 / 149.1 ; −1.65	+199.6 / 71.0 ; −5.54	−160.8 / 171.4 ; −8.56	−44.2 / 211.3 ; −2.55	−14.9 / 15.9 ; −0.59	−18.5 / 29.5 ; −2.07	−51.1 / 52.0 ; −1.66	−110.0 / 113.7 ; −5.55
AVAXUSDC → AVAXUSDT	−80.1 / 124.0 ; −0.92	−144.4 / 188.7 ; −0.62	−9.9 / 52.1 ; −1.59	+68.7 / 80.7 ; −3.36	+18.3 / 5.6 ; −3.43	+18.5 / 5.3 ; −4.85	−13.7 / 50.2 ; −1.09	−22.0 / 49.1 ; −0.60

Each cell: crash-day return / crash-day max drawdown (%) ; chained return over the 53 forward sessions without the crash (%).

None of them bounds the crash. A circuit breaker pauses, then resumes into a market still falling; the cap fixes stop the quoter buying more but leave it holding a full position through an 80% fall. The daily loss limit comes closest, and fails for a reason in its own definition: at the limit it makes quoting *reduce-only*, and a passive reducing quote does not fill while the price runs away from it.

That suggested the hypothesis, tested next: **a stop** — at the limit, send the whole position out at the opposite touch (an immediate-or-cancel order, taker fee paid), once a second until flat (`loss_flatten` in the kernel). Five variants ran on every session:

configuration	as frozen	stop 50 bp	stop 100 bp	trailing stop 100 bp, 30 min	units cap + stop 100 bp	units cap + inventory flatten 0.5
XRPUSDC → XRPPERP	+3.6 / 46.4 ; +3.96	−0.6 / 2.7 ; +4.18	−1.2 / 3.2 ; +4.23	+0.1 / 2.3 ; +4.94	−1.1 / 3.0 ; +4.23	+17.4 / 14.5 ; +3.96
XRPUSDT → XRPUSDT	+34.7 / 7.7 ; −0.09	−0.6 / 0.8 ; +0.47	+34.7 / 7.7 ; −2.60	+2.7 / 2.4 ; +0.07	+15.6 / 3.4 ; −2.58	−0.2 / 10.2 ; −2.18
SUIUSDC → SUIPERP	−4.3 / 8.9 ; −3.94	−2.7 / 6.4 ; −1.35	−1.8 / 5.5 ; −7.24	+0.4 / 3.3 ; −6.21	−1.8 / 5.5 ; −7.24	−3.1 / 8.6 ; −6.92
SUIUSDC → SUIUSDT	+19.0 / 35.1 ; −5.79	−0.5 / 1.0 ; −3.21	−1.0 / 1.8 ; −5.39	+2.3 / 1.5 ; −11.59	−1.0 / 1.8 ; −4.59	−37.9 / 46.9 ; −4.37
SUIUSDT → SUIUSDT	−143.2 / 149.1 ; −1.65	−0.5 / 0.8 ; −1.32	−1.0 / 1.4 ; −6.16	−2.5 / 3.0 ; −2.51	−1.0 / 1.4 ; −6.17	−51.1 / 52.0 ; +0.44
AVAXUSDC → AVAXUSDT	−80.1 / 124.0 ; −0.92	−0.6 / 3.6 ; −4.37	−1.1 / 4.1 ; −4.43	+4.8 / 2.1 ; −5.14	−1.0 / 4.0 ; −4.51	−17.5 / 43.8 ; −1.53

As above, for the stops.

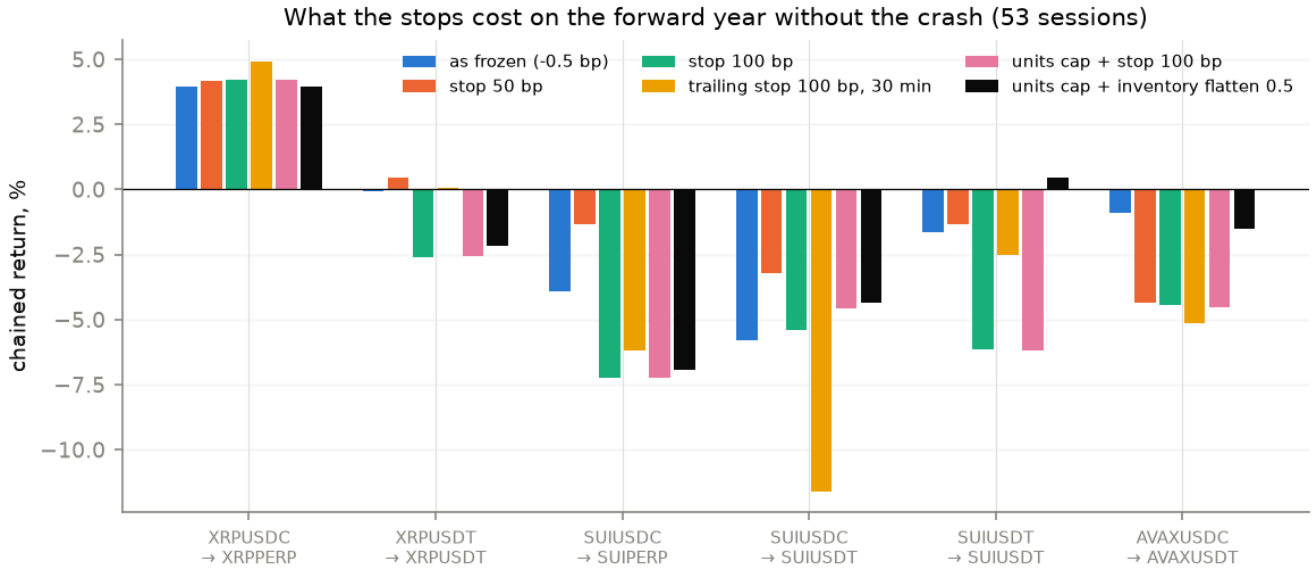


Figure — the forward year without the crash, as frozen and with each stop.

The selection rule, fixed before the stops ran: every configuration's crash-day drawdown at most 10%, then the best summed return over the forward year without the crash.

variant	worst crash-day drawdown, %	passes	summed forward return without the crash, %	median fills/day, per configuration
as frozen	149.1	no	−8.41	42, 206, 28, 88, 32, 46
breaker vol 2.5×, 30 s	188.7	no	−7.36	42, 205, 21, 76, 25, 33
breaker vol 4×, 5 min	171.4	no	−5.39	41, 236, 21, 68, 26, 36
breaker flow 0.7, hit side, 5 min	211.3	no	−17.54	30, 149, 25, 70, 19, 23
daily loss limit 50 bp	26.4	no	−7.81	42, 204, 21, 81, 28, 43
daily loss limit 100 bp	29.5	no	−15.12	42, 206, 26, 84, 28, 44
units cap, resting counted	52.0	no	−7.21	42, 210, 28, 88, 32, 46
units cap + breaker vol 2.5×	113.7	no	−7.08	42, 205, 21, 76, 25, 33
stop 50 bp	**6.4**	**yes**	**−5.59**	42, 194, 19, 70, 22, 33
stop 100 bp	7.7	yes	−21.57	42, 206, 20, 80, 24, 43
trailing stop 100 bp, 30 min	3.3	yes	−20.44	43, 206, 31, 86, 29, 44
units cap + stop 100 bp	5.5	yes	−20.85	42, 210, 20, 80, 24, 43
units cap + inventory flatten 0.5	52.0	no	−10.60	42, 217, 28, 87, 29, 45

The rule picks **stop 50 bp**. It also helps on ordinary days: by cutting a losing session at −0.5% it turns XRPUSDT's −0.09% into +0.47% and halves the SUI losses, at the price of AVAXUSDT (−0.92 → −4.37%), where sessions that dipped and recovered are now stopped out. The wider stops pay that price more often than they save it: a 100 bp stop is hit late, after the damage, and then still sells at the bottom. It does not make the five losing configurations profitable; it makes their losses bounded.

6. The picked stop, stress-tested

The picked stop, run with each of the forward test's stresses on all 61 sessions (`C10_VARSET=stopstress`):

configuration (stop 50 bp)	as picked	fee 0	fee +1 bp	fee +2 bp (Bybit VIP 0)	latency ×4	queue risk-averse	fills/day
XRPUSDC → XRPPERP	**+4.18**	**+3.45**	**+1.47**	**+0.32**	**+2.09**	**+3.56**	42
XRPUSDT → XRPUSDT	+0.47	−3.29	−11.84	−15.28	−2.18	−0.03	194
SUIUSDC → SUIPERP	−1.35	−1.78	−2.57	−4.96	−10.74	−0.73	19
SUIUSDC → SUIUSDT	−3.21	−4.81	−7.28	−8.76	−7.19	−2.30	70
SUIUSDT → SUIUSDT	−1.32	−1.53	−2.09	−3.50	−2.38	−1.40	22
AVAXUSDC → AVAXUSDT	−4.37	−4.82	−5.42	−7.23	−2.76	−4.60	33
worst crash-day drawdown, %	6.4	6.4	6.2	6.0	3.6	3.5	

Chained return, %, over the 53 forward sessions without the crash; the last row is the worst of the six configurations' drawdowns on 2025-10-10 under each stress.

- **The stop's protection is robust.** Under every stress the crash-day drawdown stays at or below 6.4% for every configuration: slower orders (latency ×4) and a pessimistic queue do not let the position run away before the stop fires.
- **Its ordinary-day effect is not.** Only XRPUSDC → XRPPERP stays positive under every stress, and the stop makes it a little sturdier than as frozen (+0.32% against +0.21% at +2 bp, +2.09% against +0.52% at latency ×4). XRPUSDT's +0.47% is the rebate: at a zero fee it is −3.29%. The other four lose under every stress.
- So the stop is a **risk control, not an edge**: it is what any of these quoters would need before running live, and it does not make one profitable that was not.

7. Lead-lag: forecasts with outside data

For ETHUSDC, XRPUSDC and SUIUSDC (C9's assets, on the study month's Binance books), four free outside sources of the same asset (`external.py`): Binance's USDT perpetual and USDT spot trades (`data.binance.vision`), Bybit's USDT perpetual trades and its book's mid. Each is reduced to a price and signed-notional series; an event counts only once it would have arrived — 5 ms after its own timestamp for Binance's markets, 80 ms for Bybit's (Singapore to Tokyo plus its publishing delay) — plus a stress of +100 or +250 ms. From each source, per 100 ms step (`leadlag.py`): its returns over 0.1, 0.5, 1 and 5 s; its signed flow over the same windows; its basis to our mid and that basis's deviation from its minute average; and its trade burst (trades in the last second over the 10-minute rate): 39 features beside C9's 17 own-book ones.

Same models and settings as C9 (boosted trees), fitted on the 24 training sessions, scored on the 7 held-out ones: the 1 s and 3 s mid move, and the large-move classifier (will the mid move against a side by the training sessions' top-1% 3 s move?).

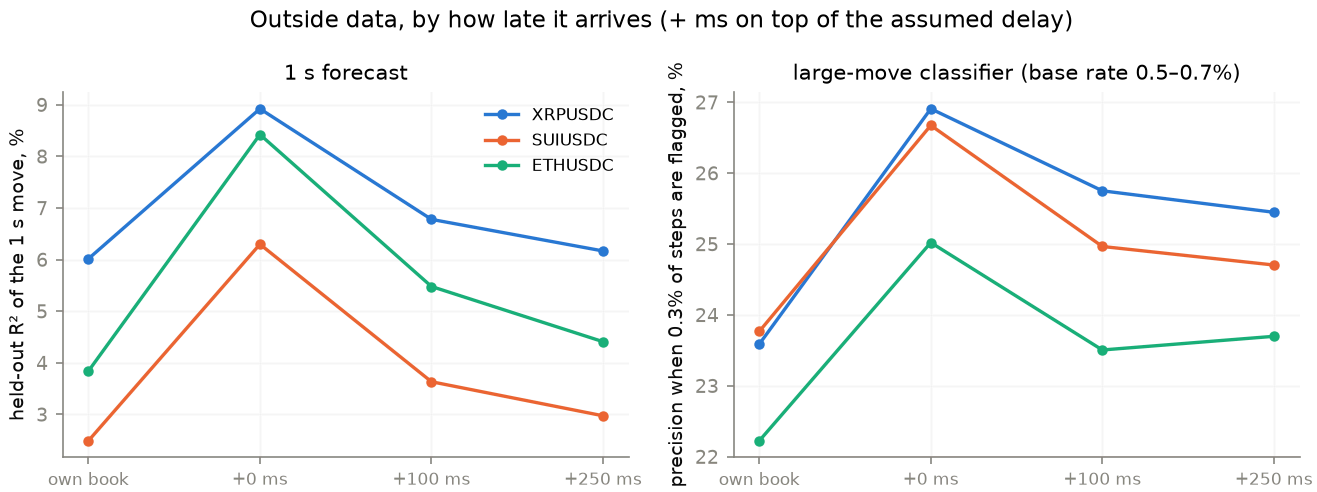


Figure — held-out R² of the 1 s move and the classifier's precision when 0.3% of steps are flagged, from the own book alone and with outside data arriving +0, +100 and +250 ms late.

asset	inputs	R² 1 s, %	R² 3 s, %	large-move AUC	precision at 0.3% flagged, %	precision at 1% flagged, %
XRPUSDC (base rate 0.69%)	own book	6.01	2.90	0.839	23.6	13.6
	+ outside, +0 ms	**8.92**	**4.11**	**0.855**	**26.9**	**15.2**
	+ outside, +100 ms	6.78	3.71	0.853	25.8	14.8
	+ outside, +250 ms	6.17	3.09	0.852	25.4	14.8
SUIUSDC (base rate 0.70%)	own book	2.49	1.38	0.786	23.8	14.1
	+ outside, +0 ms	**6.30**	**2.96**	**0.807**	**26.7**	**15.1**
	+ outside, +100 ms	3.63	1.82	0.801	25.0	14.4
	+ outside, +250 ms	2.97	1.56	0.802	24.7	14.4
ETHUSDC (base rate 0.50%)	own book	3.84	1.83	0.855	22.2	12.3
	+ outside, +0 ms	**8.42**	**3.72**	**0.871**	**25.0**	**13.7**
	+ outside, +100 ms	5.48	2.29	0.869	23.5	13.4
	+ outside, +250 ms	4.41	1.99	0.870	23.7	13.3

- **The lead is real and short.** With the data as assumed, outside markets add 3–5 points of R² to the 1 s forecast — a 50–150% gain. 100 ms later half of it is gone, 250 ms later almost all. The informative features are Binance spot's 5 s return and the trade bursts

on Binance spot and Bybit's perpetual: someone trades there first, and our book follows within a few hundred milliseconds.
- **Large moves are not much better anticipated.** AUC rises by 0.016–0.021 and precision at 0.3% flagged by 2–3 points, and that small gain survives the delay. Three in four alarms are still false.
- **Still far from what pays.** C9 found a forecast has to explain about 40% of the 1 s move before quoting around it beats the C8 choice; the best here is 8.9%.

8. Lead-lag through the backtest

Pending: `llbacktest.py` *— each forecast (own book, own book + outside) used as a fair-price shift and as a pull rule on the C8 OBI configurations, chosen on training sessions, reported held out and stress-tested including +100/+250 ms; running on Kaggle.*

9. So what?

To be written when §6 and §8 are in.

10. How it was run, and what went wrong

- **Where.** Bybit conversions and replays of the two small USDC contracts on this machine (4 cores, 16 GB) and on Kaggle CPU sessions; the three USDT contracts (12–48 million book events a day, up to 12 GB to convert) on Kaggle only, sharded 12 ways. The lead-lag fits on Kaggle (37 minutes a symbol including the downloads). About 6,600 Bybit session replays in all, before the stop's stress tests.
- **The container restarted every 30–60 minutes**, killing local work. Every unit is cached as it finishes and every fit saves model by model, so a restart costs one unit; heavy work moved to Kaggle.
- **A truncated download** (the connection dropped and the file was cached as if whole) broke one SUIUSDT day; downloads now check Content-Length.
- **A converter that ran out of memory** on the largest days now builds its event columns in 2-million-row chunks.
- **Bybit's REST API is region-blocked here**, so tick and lot sizes are measured from the data.